








Advanced Time Series Analysis and Forecasting

Seminar 10: Long memory and rough volatility



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- **CAC**: Cotation Assistée en Continu (CAC 40 index, Paris) — the CAC 40 index of the Paris bourse
- **CEE**: Central and Eastern Europe
- **DM**: Diebold–Mariano (test)
- **ELW**: Exact Local Whittle (estimator)
- **EODHD**: EOD Historical Data (market-data provider)
- **FCVAR**: Fractionally Cointegrated VAR
- **FFT**: Fast Fourier Transform
- **FIGARCH**: Fractionally Integrated GARCH

Acronyms used in this material (2/2)

- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GPH**: Geweke–Porter-Hudak (estimator)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **LW**: Local Whittle (estimator)
- **MSE**: Mean Squared Error
- **OLS**: Ordinary Least Squares
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- **QLIKE**: Quasi-LIKElihood loss
- **RFSV**: Rough Fractional Stochastic Volatility (model)
- **RV**: Realised Variance
- **SD**: Standard Deviation
- **SE**: Standard Error
- **VECM**: Vector Error Correction Model
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and seminar structure

- The question of the seminar
  - **how do we measure, test and use long memory and roughness in volatility and inflation?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	ARFIMA autocorrelations and the mean; fractional weights	Solved, Proposed	A1
A3, A4	standard errors of GPH and LW; bias and bandwidth	Solved, Proposed	A3
A5, A6	fractional Gaussian noise; the RFSV kernel	Solved, Proposed	A5
A7, A8	circulant embedding by hand; regime switching and spurious long memory	Solved, Proposed	A7, A1
B1, B2	memory of FTSE 100 log RV; BET and EUR/RON with the Qu test	Solved, Proposed	B1
B3, B4	roughness of the Euro Stoxx 50; Bitcoin by sampling interval	Solved, Proposed	B3
B5, B6	forecasting RV of the FTSE 100 and the CAC 40	Solved, Proposed	B5
C1, C2	fractional cointegration of two volatilities; what is wrong in an AI answer?	Proposed	B1, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar
- Part A: derivations and computations on paper; Part B: real data, each exercise ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
RV of FTSE 100, Euro Stoxx 50, DAX, CAC 40	Heber et al. (2009)	daily, 5-minute returns	2000–2022
RV of Bitcoin	Binance one-minute prices	daily, 1–60 minutes	2018–2026
BET	EODHD	daily	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026

- RV in $\%^2$, returns in %
- Functions in the notebook
  - `gph`, `local_whittle`, `elw`, `qu_stat`, `gjr_scaling`, `h_with_noise`, `rfsv_kernel`, `forecast_rv`, `fcvar_fit`
- No account or key is needed

Prerequisites for Today (1/24): autocorrelation and memory

- **Autocovariance** and **autocorrelation** of a stationary series X_t at lag k

$$\gamma(k) = \mathrm{Cov}(X_t, X_{t+k}), \qquad \rho(k) = \frac{\gamma(k)}{\gamma(0)} \in [-1, 1]$$

  - $\gamma(0) = \mathrm{Var} X_t$; stationary: mean and $\gamma(k)$ do not depend on t

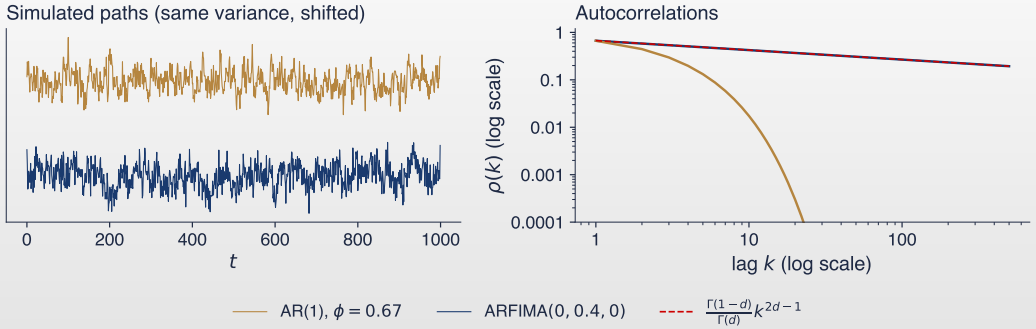
- **Short memory**: autocorrelations decay exponentially and are summable
  - AR(1) $X_t = \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$: $\rho(k) = \phi^k$, so $\sum_k |\rho(k)| < \infty$
  - ε_t: white noise (uncorrelated, mean 0, variance σ^2)

- **Long memory**: hyperbolic decay, not summable, a spectral pole at frequency zero

$$\rho(k) \sim c\, k^{2d-1} \ (k \to \infty), \qquad f(\lambda) \sim G \lambda^{-2d} \ (\lambda \to 0), \qquad 0 < d < 1/2$$

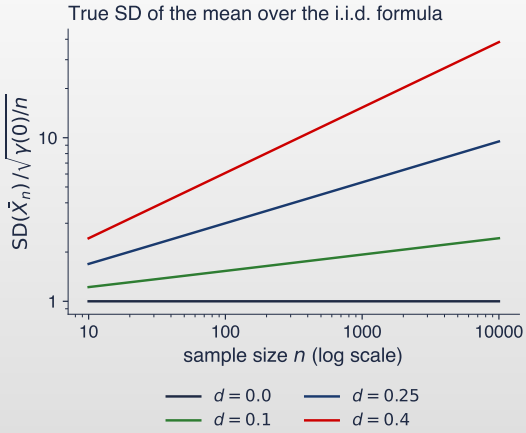
  - d: memory parameter; $c, G > 0$: constants; $f(\lambda)$: spectral density at frequency λ (defined below)
  - $a_k \sim b_k$: the ratio $a_k / b_k \to 1$; Hurst exponent $H = d + 1/2$

Prerequisites for Today (2/24): hyperbolic against exponential decay



- ARFIMA$(0, 0.4, 0)$ (defined below) and an AR(1) with the same $\rho(1)$, $\phi = 0.67$; both simulated, $n = 1000$
- At lag 100: $\rho = 0.27$ for ARFIMA, practically 0 for the AR(1); on log scales the hyperbolic ACF is a straight line of slope $2d - 1$
- The ARFIMA path wanders in long swings around its mean

Prerequisites for Today (3/24): the variance of the sample mean



- For any stationary series

$$\mathrm{Var}(\bar{X}_n) = \frac{1}{n} \sum_{|k|<n} \Big(1 - \frac{|k|}{n}\Big)\gamma(k)$$

  - $\bar{X}_n = \frac{1}{n}\sum_{t=1}^{n} X_t$; n: sample size
  - i.i.d.: only $k = 0$ remains, $\gamma(0)/n$

- Long memory: of order n^{2d-1}, not n^{-1}
  - the ratio in the chart grows like n^d
  - $n = 1000$: 1.9, 5.3, 15.3 for $d = 0.1$, 0.25, 0.4
  - the i.i.d. standard error is that many times too small

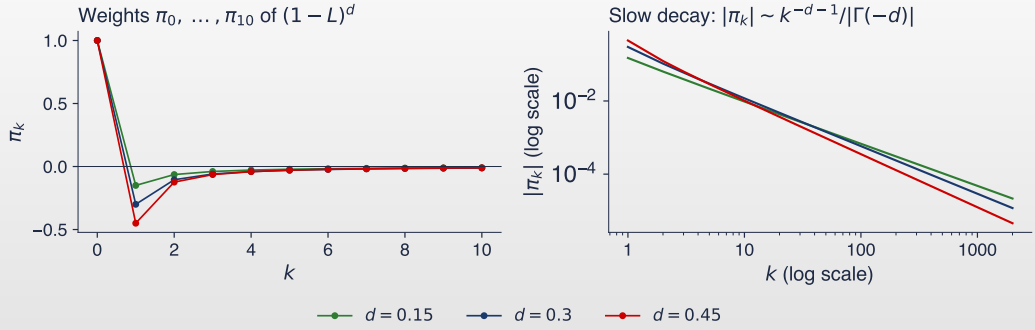
Prerequisites for Today (4/24): the lag operator and fractional differencing

- **Lag operator** L: $LX_t = X_{t-1}$, $L^k X_t = X_{t-k}$; first difference $(1-L)X_t = X_t - X_{t-1}$
- **Fractional difference**: the binomial series of $(1-L)^d$ for a real d

$$(1-L)^d X_t = \sum_{k=0}^{\infty} \pi_k X_{t-k}, \qquad \pi_0 = 1, \quad \pi_k = \pi_{k-1} \frac{k-1-d}{k}$$

  - π_k: weight of X_{t-k}; $d = 1$: $\pi_1 = -1$, $\pi_k = 0$ for $k \geq 2$ (the usual difference)
  - $0 < d < 1$: all past values enter, with weights $\pi_k \sim k^{-d-1}/\Gamma(-d)$
  - $\Gamma(z) = \int_0^\infty u^{z-1} e^{-u} du$: gamma function, $\Gamma(n) = (n-1)!$; $\Gamma(-d) < 0$ for $0 < d < 1$
  - $\sum_k \pi_k = (1-1)^d = 0$: a constant level is removed only by the infinite filter

Prerequisites for Today (5/24): fractional weights



- Left: $\pi_{\mathbf{1}} = -d$, then small negative weights that shrink slowly
- Right: on log scales $|\pi_k|$ is a straight line of slope $-(d+1)$: a hyperbolic, not exponential, decay
- A filter truncated after K lags leaves out the tail of the weights

Prerequisites for Today (6/24): the ARFIMA model

- **ARFIMA**$(0, d, 0)$ Hosking (1981): white noise after fractional differencing

$$(1 - L)^d X_t = \varepsilon_t, \qquad -\tfrac{1}{2} < d < \tfrac{1}{2} \text{ (stationary and invertible)}$$

  - $d > 0$: long memory; $d < 0$: antipersistence ($\sum_k \rho(k) = 0$); $d \geq 1/2$: nonstationary
- Autocorrelations: a recursion and its limit

$$\rho(0) = 1, \quad \rho(k) = \rho(k-1)\frac{k - 1 + d}{k - d}, \qquad \rho(k) \sim \frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$$

- **ARFIMA**$(1, d, 0)$: $(1 - \phi L)(1 - L)^d X_t = \varepsilon_t$
  - ϕ: AR coefficient, $|\phi| < 1$; it changes the short lags, d the long ones
  - ARFIMA(p, d, q): an ARMA(p, q) for $(1 - L)^d X_t$

Prerequisites for Today (7/24): the spectral density and the periodogram

- **Spectral density**: the variance of X_t decomposed by frequency

$$f(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma(k) e^{-i\lambda k}, \qquad \gamma(0) = \int_{-\pi}^{\pi} f(\lambda)\, d\lambda$$

  - $\lambda \in [-\pi, \pi]$: angular frequency, a cycle of $2\pi/\lambda$ periods; low λ: long cycles
  - ARFIMA$(0, d, 0)$: $f(\lambda) = \frac{\sigma^2}{2\pi} \left(4 \sin^2(\lambda/2)\right)^{-d} \sim \frac{\sigma^2}{2\pi} \lambda^{-2d}$
- **Periodogram** at the Fourier frequencies: the sample counterpart of f

$$I(\lambda_j) = \frac{1}{2\pi n} \left| \sum_{t=1}^{n} X_t e^{-i\lambda_j t} \right|^2, \qquad \lambda_j = \frac{2\pi j}{n}, \quad j = 1, \ldots, \lfloor n/2 \rfloor$$

  - $I(\lambda_j)/f(\lambda_j)$ is approximately exponential with mean 1, independent across j: very noisy
  - bandwidth m: the number of lowest frequencies $\lambda_1, \ldots, \lambda_m$ used to estimate d; $m \to \infty$, $m/n \to 0$

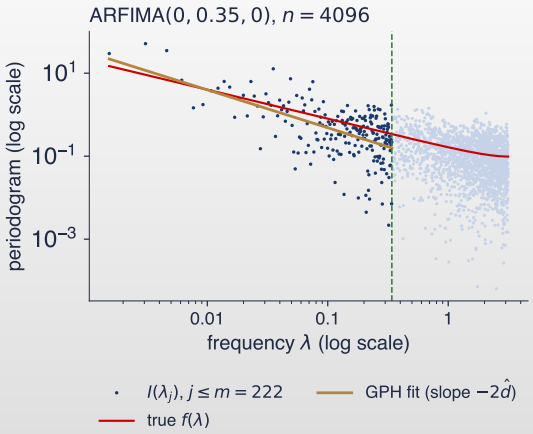
Prerequisites for Today (8/24): the GPH estimator

- **GPH** Geweke and Porter-Hudak (1983): take logs of $I(\lambda_j) \approx f(\lambda_j)\,\xi_j$ near zero and run OLS

$$\log I(\lambda_j) = a + d\left[-\log\left(4\sin^2(\lambda_j/2)\right)\right] + u_j, \qquad j = 1, \ldots, m$$

  - ξ_j: exponential error with mean 1; $u_j = \log \xi_j - \mathbb{E}\log \xi_j$; a: intercept
  - $\hat{d}$: the OLS slope; regressor $\approx -2\log \lambda_j$, so the log-log slope is $-2d$
- Asymptotic standard error
  - $\mathrm{Var} u_j = \pi^2/6$ and the regressor has variance ≈ 4: $\mathrm{SE}(\hat{d}) = \pi/\sqrt{24m}$
  - $\sqrt{m}(\hat{d} - d) \to N(0, \pi^2/24)$; 95% interval $\hat{d} \pm 1.96\,\mathrm{SE}$
- Precision grows only with $\sqrt{m}$, not with $\sqrt{n}$: estimating d needs long samples

Prerequisites for Today (9/24): the log-periodogram near zero



- Simulated ARFIMA$(0, 0.35, 0)$, $n = 4096$
- The $m = n^{0.65} = 222$ lowest ordinates (dark) lie around a line
  - GPH: $\hat{d} = 0.46$; local Whittle: 0.44
  - both within two SEs of $d = 0.35$
- The scatter around the line is the exponential noise ξ_j: it does not shrink with n
- Far from zero (pale points) the short-run dynamics bend f: they must stay out of the regression

Prerequisites for Today (10/24): the local Whittle estimator

- **Local Whittle** Robinson (1995b): Gaussian likelihood of $f(\lambda_j) = G\lambda_j^{-2d}$ on the m lowest frequencies

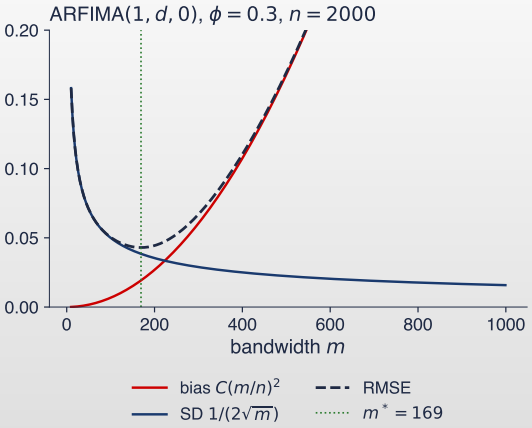
$$\hat{d} = \arg\min_d R(d), \qquad R(d) = \log \hat{G}(d) - \frac{2d}{m}\sum_{j=1}^{m} \log \lambda_j, \quad \hat{G}(d) = \frac{1}{m}\sum_{j=1}^{m} \lambda_j^{2d} I(\lambda_j)$$

  - $\hat{G}(d)$: the level G that fits best for a given d; $R(d)$: the profiled objective (a QLIKE-type fit)

- Inference
  - $\sqrt{m}(\hat{d} - d) \to N(0, 1/4)$: $\mathrm{SE} = 1/(2\sqrt{m})$, smaller than GPH at the same m
  - relative efficiency: the ratio of the two asymptotic variances, $(\pi^2/24)/(1/4)$
  - consistent for $-1/2 < d \leq 1$; for $d > 1$ it converges to 1

- **Exact local Whittle** (ELW) Shimotsu and Phillips (2005), Shimotsu (2010)
  - replaces $\lambda_j^{2d} I(\lambda_j)$ by the periodogram of $(1-L)^d(X_t - \hat{\mu})$, $\hat{\mu}$: estimated mean
  - $N(d, 1/(4m))$ for any d in the search range, also $d \geq 1/2$ (nonstationary)

Prerequisites for Today (11/24): bias and the choice of bandwidth



- Short-run dynamics bias $\hat{d}$; for ARFIMA$(1, d, 0)$

$$\mathbb{E}\hat{d} - d \approx C\Big(\frac{m}{n}\Big)^{\mathbf{2}}, \quad C = \frac{2\pi^{\mathbf{2}}}{9} \cdot \frac{2\phi}{(1-\phi)^{\mathbf{2}}}$$

  - $\phi > 0$: positive bias, growing with m
- MSE $=$ bias$^{\mathbf{2}} + \frac{\mathbf{1}}{\mathbf{4}m}$; minimised at

$$m^* = \Big(\frac{n^{\mathbf{4}}}{16C^{\mathbf{2}}}\Big)^{\mathbf{1/5}}$$

  - MSE: mean squared error; m^* grows like $n^{\mathbf{4/5}}$
  - chart: $\phi = 0.3$, $n = 2000$: $C = 2.69$, $m^* = 169 = n^{\mathbf{0.68}}$

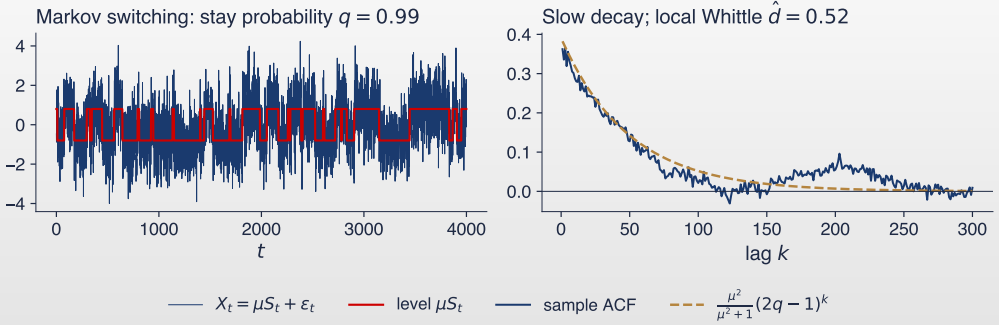
Prerequisites for Today (12/24): level shifts that mimic long memory

- **Two-state Markov chain** $S_t \in \{-1, 1\}$: it stays with probability q, switches with $1 - q$

$$X_t = \mu S_t + \varepsilon_t, \qquad \mathbb{E}(S_t S_{t+k}) = (2q - 1)^k$$

  - μ: size of the level; ε_t: i.i.d. $N(0, 1)$, independent of S; expected duration of a regime $1/(1 - q)$
  - the ACF decays exponentially: short memory in the strict sense
- **Half-life** of a geometric decay $\rho(k) = \rho(0)\, r^k$
  - $k_{1/2} = \ln 0.5 / \ln r$: the lag at which the autocorrelation has halved
- Rare shifts, structural breaks and slowly varying trends put power at the lowest frequencies Diebold and Inoue (2001)
  - $\hat{d}$ is then large although the process has short memory: **spurious long memory**

Prerequisites for Today (13/24): a simulated switching series



- $q = 0.99$ (regimes last 100 periods on average), $\mu = 0.8$, $n = 4000$
- Local Whittle with $m = n^{\mathbf{0.65}}$ gives $\hat{d} = 0.52$, a value typical of a long-memory series
- Over a few hundred lags the slow geometric decay cannot be told apart from a hyperbolic one

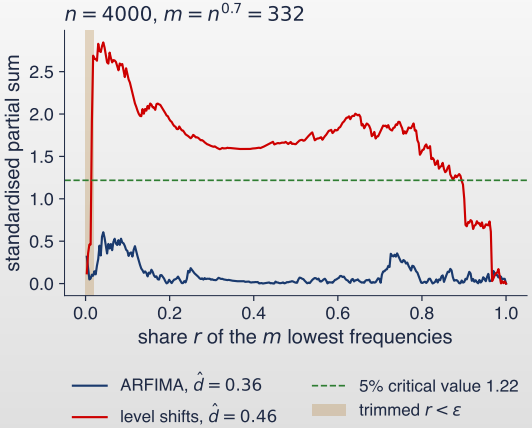
Prerequisites for Today (14/24): the Qu test

- **Qu test** Qu (2011): H_0 stationary long memory, $f(\lambda) = G\lambda^{-2d}$ on the m lowest frequencies

$$W = \sup_{r \in [\varepsilon, 1]} \Big(\sum_{j=1}^{m} \nu_j^2 \Big)^{-1/2} \Big| \sum_{j \le mr} \nu_j \Big(\frac{\lambda_j^{2\hat{d}} I(\lambda_j)}{\hat{G}} - 1 \Big) \Big|$$

  - $\hat{d}$, $\hat{G} = \hat{G}(\hat{d})$: local Whittle estimates; $\nu_j = \log \lambda_j - \frac{1}{m} \sum_i \log \lambda_i$
  - the bracket is the score of local Whittle at frequency j; its full sum is zero at $\hat{d}$
  - r: share of the m frequencies summed; ε: trimming, the first εm partial sums are skipped
- Reading
  - under H_0 the partial sums stay small at every r; level shifts make the lowest ordinates too large: a big W
  - reject for W above the critical value; with $m = n^{0.7}$, $\varepsilon = 0.02$ (simulated): 1.09 (10%), 1.22 (5%), 1.56 (1%)

Prerequisites for Today (15/24): the Qu partial sums



- Two simulated series, $n = 4000$
- ARFIMA$(0, 0.35, 0)$: the curve stays low
  - $W = 0.61$: do not reject
- Markov level shifts ($q = 0.997$, $\mu = 0.8$)
  - $\hat{d}$ looks like long memory
  - but the curve jumps at small r: $W = 2.85$, reject at 1%
- A rejection says only that stationary long memory does not fit: breaks, shifts or trends are alternatives

Prerequisites for Today (16/24): fractional cointegration

- **Fractional cointegration**: p series, each with memory d, whose combinations $\beta' X_t$ have memory $d - b$, $b > 0$
- **FCVAR** Johansen and Nielsen (2012): a VECM written in fractional differences

$$\Delta^d X_t = \alpha\beta' L_b \Delta^{d-b} X_t + \sum_{i=1}^{k} \Gamma_i \Delta^d L_b^i X_t + \varepsilon_t, \qquad \Delta^d = (1-L)^d, \quad L_b = 1 - \Delta^b$$

  - β ($p \times r$): cointegrating vectors; α ($p \times r$): adjustment speeds; r: cointegration rank
  - L_b: fractional lag operator (only past values); k: number of lagged short-run terms, Γ_i their matrices; $b = d = 1$: the usual VECM

- Estimation and rank tests
  - for fixed (d, b): Johansen reduced-rank regression; profile likelihood: maximise over a grid of (d, b)
  - trace test of rank r against p: $LR = 2(\ell_p - \ell_r)$, ℓ: maximised log likelihoods; $\chi^2_{(p-r)^2}$ when $b < 1/2$

Prerequisites for Today (17/24): fractional Brownian motion

- **Fractional Brownian motion** (fBm) W_t^H Mandelbrot and Van Ness (1968): Gaussian, $W_0^H = 0$, mean 0, covariance

$$\mathbb{E}(W_t^H W_s^H) = \tfrac{1}{2}\left(t^{2H} + s^{2H} - |t-s|^{2H}\right), \qquad \mathbb{E}|W_{t+\Delta}^H - W_t^H|^2 = \Delta^{2H}$$

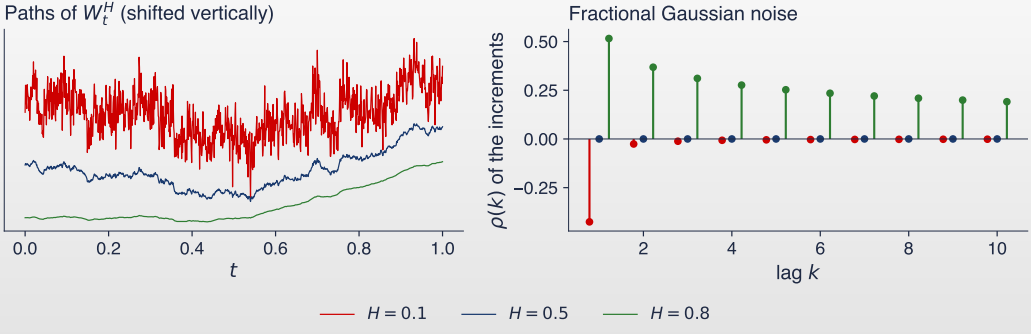
  - $H \in (0,1)$: Hurst exponent; $H = 1/2$: Brownian motion, independent increments
  - $H < 1/2$: rough (irregular) paths, negatively correlated increments; $H > 1/2$: smooth paths, persistent increments

- **Fractional Gaussian noise** (fGn): the unit increments $W_{t+1}^H - W_t^H$

$$\gamma_k = \rho(k) = \tfrac{1}{2}\left(|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H}\right)$$

  - $\rho(1) = 2^{2H-1} - 1$; for $H > 1/2$, $\rho(k) \sim H(2H-1)k^{2H-2}$: long memory with $d = H - 1/2$

Prerequisites for Today (18/24): rough and smooth paths



- Left: fBm paths from the same random numbers for $H = 0.1, 0.5, 0.8$
- Right: $H = 0.1$ gives $\rho(1) \approx -0.43$ and almost nothing beyond: the path reverts at once
- $H = 0.8$: positive, slowly decaying autocorrelations, the long memory of the previous frames

Prerequisites for Today (19/24): circulant embedding

- **Goal**: simulate n points of a stationary Gaussian series with autocovariances $\gamma_0, \ldots, \gamma_{n-1}$ exactly
  - Cholesky factorisation of the $n \times n$ covariance costs $O(n^3)$
- **Davies–Harte** Davies and Harte (1987): embed the covariance in a circulant matrix of size $M = 2(n-1)$

$$c = (\gamma_0, \gamma_1, \ldots, \gamma_{n-1}, \gamma_{n-2}, \ldots, \gamma_1), \qquad \lambda_k = \sum_{j=0}^{M-1} c_j e^{-2\pi i j k / M}$$

  - circulant: each row is the previous one shifted by one place; c: its first row
  - λ_k: its eigenvalues, the FFT (fast Fourier transform) of c, real because c is symmetric
  - if all $\lambda_k \geq 0$: $Y = \mathrm{FFT}\big(\sqrt{\lambda / M}\, Z\big)$, Z complex Gaussian; the first n values have the target covariance
  - cost $O(n \log n)$; for fGn all $\lambda_k \geq 0$ for every H

Prerequisites for Today (20/24): realised variance and moment scaling

- **Realised variance** of day t from N_t intraday returns $r_{t,i}$ (e.g. 5-minute)

$$RV_t = \sum_{i=1}^{N_t} r_{t,i}^2, \qquad \sigma_t = \sqrt{RV_t}$$

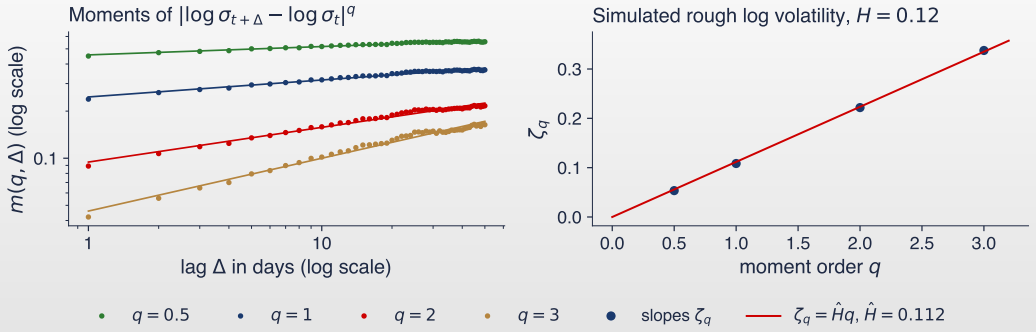
  - a noisy measure of the daily variance; fewer returns per day: a larger measurement error

- **Moment scaling** Gatheral, Jaisson and Rosenbaum (2018): for $\log \sigma_t = \nu W_t^H$ the moments of the increments are power laws

$$m(q, \Delta) = \frac{1}{N} \sum_t |\log \sigma_{t+\Delta} - \log \sigma_t|^q \propto \Delta^{\zeta_q}, \qquad \zeta_q = qH$$

  - q: moment order; Δ: lag in days; N: number of pairs; ν: volatility of volatility (vol-of-vol)
  - ζ_q: slope of $\log m(q, \Delta)$ on $\log \Delta$; then $\hat{H} = \sum_q q\zeta_q / \sum_q q^2$ (regression through the origin)
  - i.i.d. measurement error of variance s^2 adds an intercept: $m(2, \Delta) = \nu^2 \Delta^{2H} + 2s^2$

Prerequisites for Today (21/24): the scaling regression



- Simulated daily log volatility $0.3\, W_t^H$ with $H = 0.12$, 4000 days, no measurement error
- Left: straight lines on log scales, one per q; right: the slopes lie on a line through the origin, $\hat{H} = 0.112$
- Departures from linearity in q (multifractality, heavy-tailed increments) show up at large q

Prerequisites for Today (22/24): HAR, FIGARCH and lognormal forecasts

- **HAR** Corsi (2009): daily, weekly and monthly averages of the past, $y_t = \log RV_t$

$$y_{t+h} = \beta_0 + \beta_d y_t + \beta_w \bar{y}_t^{(5)} + \beta_m \bar{y}_t^{(22)} + u_{t+h}, \qquad \bar{y}_t^{(K)} = \frac{1}{K} \sum_{j=0}^{K-1} y_{t-j}$$

  - h: horizon; OLS; a short-memory AR(22) with restricted weights that mimics long memory

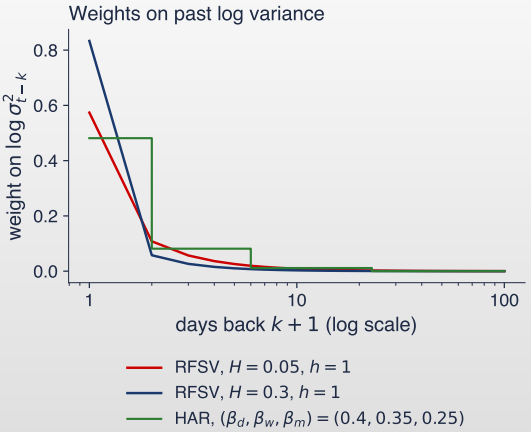
- **FIGARCH** Baillie, Bollerslev and Mikkelsen (1996): the GARCH variance with a fractional difference of the squared shocks e_t^2

$$\sigma_t^2 = \omega + \left[1 - (1 - \beta L)^{-1}(1 - \varphi L)(1 - L)^d\right] e_t^2$$

  - $\omega > 0$, β, φ: GARCH-type coefficients; the bracket is an ARCH(∞) filter whose weights sum to one

- **Lognormal correction**: models forecast $\log RV$, the target is RV
  - if $\log RV_{t+h} \sim N(\mu, s^2)$: $\mathbb{E}RV_{t+h} = e^{\mu + s^2/2}$; e^{μ} alone is the median, too low

Prerequisites for Today (23/24): the RFSV predictor



- RFSV: $\log \sigma_t = \nu W_t^H$; the forecast is a power-kernel average of the past

$$\widehat{\log \sigma^2_{t+h}} = \sum_k w_k \log \sigma^2_{t-k}$$

  - $w_k \propto \int_k^{k+1} \frac{du}{(u+h)u^{H+1/2}}$, normalised to sum to 1; h: horizon

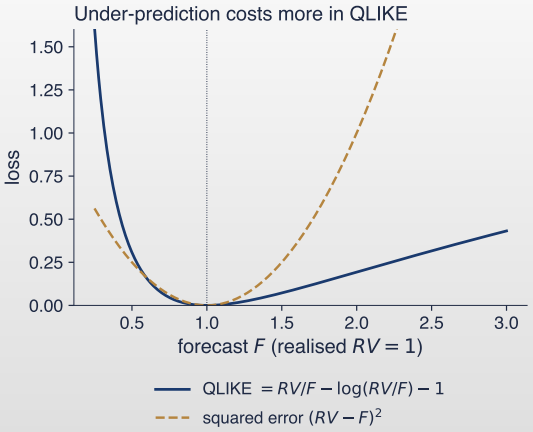
- Variance forecast

$$\exp\left(\widehat{\log \sigma^2_{t+h}} + 2c\nu^2 h^{2H}\right)$$

  - $c = \frac{\Gamma(3/2-H)}{\Gamma(H+1/2)\Gamma(2-2H)}$; $4c\nu^2 h^{2H}$: the forecast-error variance of $\log \sigma^2$, halved as in the lognormal correction

- Chart: the kernel puts most weight on the last day and a long thin tail behind; HAR uses three flat steps

Prerequisites for Today (24/24): QLIKE and the Diebold–Mariano test



- **QLIKE** Patton (2011) of a variance forecast F

$$\mathrm{QLIKE} = \frac{RV}{F} - \log\frac{RV}{F} - 1 \geq 0$$

  - 0 only if $F = RV$; ranks forecasts correctly although RV is a noisy proxy
  - penalises under-prediction more (chart)

- **DM test** (Chapter 1): δ_t = loss of model 1 minus loss of model 2

$$DM = \frac{\bar{\delta}}{\sqrt{\hat{\Omega}/T}} \to N(0,1)$$

  - $\bar{\delta}$: mean over T forecasts; $\hat{\Omega}$: Newey–West variance $\hat{\gamma}_0 + 2\sum_{k=1}^{L}(1 - \frac{k}{L+1})\hat{\gamma}_k$ of δ_t
  - $DM < -1.96$: model 1 significantly better at 5%

A1: ARFIMA autocorrelations and the mean [Solved]

Task

- ARFIMA$(0, 0.3, 0)$.
- 1. Show that $\rho(1) = d/(1-d)$ and compute $\rho(1)$, $\rho(10)$, $\rho(100)$ with the recursion.
- 2. Compare $\rho(100)$ with its asymptotic form.
- 3. For $n = 1000$, compute the ratio of the true SD of $\bar{X}$ to the i.i.d. formula $\sqrt{\gamma(0)/n}$ (notebook A1).
- Report: three numbers; two numbers; one number and one sentence.

Solution

- 1. $\rho(1) = \rho(0) \cdot \frac{0+d}{1-d}$; $\rho(1) = 0.429$, $\rho(10) = 0.173$, $\rho(100) = 0.069$
- 2. Asymptotic formula: 0.069, practically equal already at $k = 100$
- 3. Ratio 7.55: the i.i.d. standard error is 7.55 times too small
- Interpretation: with $d = 0.3$, 1000 observations carry the information of far fewer independent ones; HAC with a fixed bandwidth cannot repair this

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A2: fractional differencing weights [Proposed]

Solution discussed in class

Task

- $d = 0.4$. Model: A1.
- 1. Compute $\pi_1, \ldots, \pi_5$ of $(1 - L)^{0.4}$.
- 2. Compare π_{100} with its asymptotic form.
- 3. Compute $\sum_{k=0}^{1000} \pi_k$; what is its limit, and what does the gap mean for a truncated filter?
- Report: five numbers; two numbers; one number and one sentence.

A3: standard errors of GPH and local Whittle [Solved]

Task

- $n = 5000$, $m = \lfloor n^{0.65} \rfloor$; a local Whittle estimate $\hat{d} = 0.45$.
- 1. Compute m and the asymptotic SE of GPH and of local Whittle.
- 2. Give the 95% interval for d and the relative efficiency of LW against GPH.
- 3. Which m gives a half-width of 0.05?
- Report: three numbers; an interval and one number; one number.

Solution

- 1. $m = 253$; GPH $\pi/\sqrt{24m} = 0.0403$; LW $1/(2\sqrt{m}) = 0.0314$
- 2. $[0.388, 0.512]$; efficiency $(\pi^2/24)/(1/4) = \pi^2/6 = 1.645$
- 3. $1.96/(2\sqrt{m}) \leq 0.05 \Rightarrow m \geq 385$
- Interpretation: the interval includes $1/2$, so stationarity is not established; precision is bought with bandwidth, and bandwidth costs bias (A4)

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A4: bias and the optimal bandwidth [Proposed]

Solution discussed in class

Task

- ARFIMA$(1, d, 0)$ with $\phi = 0.5$, $n = 5000$. Model: A3.
- 1. Compute the bias constant C.
- 2. For $m = \lfloor n^{\mathbf{0.8}} \rfloor$, compute the bias and the SD of local Whittle.
- 3. Compute m^*, its exponent $\log m^* / \log n$, and the bias and SD at m^*.
- Report: one number; two numbers; four numbers and one sentence.

A5: fractional Gaussian noise and scaling [Solved]

Task

- fGn with $H = 0.1$, 0.3 and 0.7.
- 1. Compute the fGn autocorrelations $\rho(1)$ and $\rho(2)$ for each H.
- 2. Compute $m(2, 22)/m(2, 1) = 22^{2H}$: how much more does log volatility move over a month than over a day?
- 3. Show that $\zeta_q = qH$ for fBm.
- Report: six numbers; three numbers; one derivation.

Solution

- 1. $H = 0.1$: -0.426, -0.026; $H = 0.3$: -0.242, -0.049; $H = 0.7$: 0.320, 0.189
- 2. 1.86, 6.39, 75.75: with $H = 0.1$ the monthly variance of changes is less than twice the daily one
- 3. $W^H_{t+\Delta} - W^H_t \sim N(0, \Delta^{2H})$: $\mathbb{E}|\cdot|^q = \Delta^{qH}\mathbb{E}|Z|^q$; take logs
- Interpretation: rough log volatility moves a lot from day to day but drifts little: strong negative correlation of daily changes

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A6: the RFSV kernel and the variance correction [Proposed]

Solution discussed in class

Task

- $H = 0.1$, $\nu = 0.3$, horizon $h = 1$; kernel truncated at 500 days. Model: A5.
- 1. Compute the share of the weight on the last day, the last 5 and the last 22 days (notebook A6).
- 2. Compute $c(H)$ and the variance correction for $h = 1, 5, 22$.
- 3. By how much would ignoring the correction understate the variance forecast at $h = 22$?
- Report: three shares; four numbers; one number.

A7: circulant embedding by hand [Solved]

Task

- Three points of fGn with $H = 0.3$: $\gamma_0, \gamma_1, \gamma_2$.
- 1. Compute γ_1 and γ_2.
- 2. Write the first row of the 4×4 circulant and its eigenvalues λ_k ($M = 4$).
- 3. Can the Davies–Harte method be used?
- Report: two numbers; a row and four numbers; one sentence.

Solution

- 1. $\gamma_1 = \frac{1}{2}(2^{0.6} - 2) = -0.2421$,
  $\gamma_2 = \frac{1}{2}(3^{0.6} - 2 \cdot 2^{0.6} + 1) = -0.0491$
- 2. $c = (\gamma_0, \gamma_1, \gamma_2, \gamma_1)$; $\lambda_0 = \gamma_0 + 2\gamma_1 + \gamma_2 = 0.4666$,
  $\lambda_1 = \lambda_3 = \gamma_0 - \gamma_2 = 1.0491$, $\lambda_2 = \gamma_0 - 2\gamma_1 + \gamma_2 = 1.4352$
- 3. Yes: all eigenvalues are positive, so the circulant is a valid covariance and its first 3 coordinates have the fGn covariance
- Interpretation: the embedding doubles the size but turns an $O(n^3)$ factorisation into two FFTs

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A8: regime switching and spurious long memory [Proposed]

Solution discussed in class

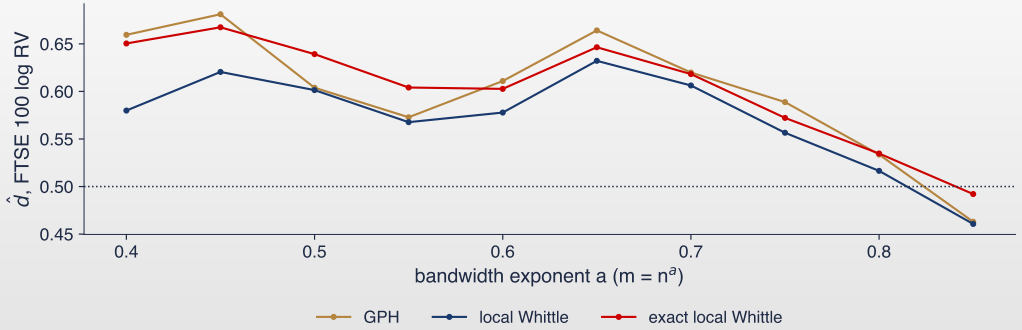
Task

- $X_t = \mu S_t + \varepsilon_t$, $S_t \in \{-1, 1\}$ a Markov chain that stays with probability $q = 0.995$, $\mu = 1$, $\varepsilon_t \sim N(0, 1)$. Model: A1.
- 1. Show that $\rho(k) = \frac{\mu^2}{\mu^2+1}(2q-1)^k$ and compute $\rho(1)$, $\rho(100)$, $\rho(500)$.
- 2. Compute the half-life of the autocorrelation.
- 3. Fit $\log \rho(k)$ on $\log k$ for $k = 5, \ldots, 200$ and convert the slope to d.
- Report: three numbers; one number; one number and one sentence.

B1: memory of FTSE 100 realised variance [Solved]

- **Question**: how large is d for the log realised variance of the FTSE 100, and is it robust to the bandwidth and to level shifts?
- **Inputs**: Oxford-Man 5-minute RV of the FTSE 100, $n = 5\,586$ days to 25 February 2022
- **Tasks**
  1. Estimate d of $\log RV$ by GPH, local Whittle and exact local Whittle for $m = n^{0.5}, n^{0.6}, n^{0.7}$, with standard errors.
  2. Plot $\hat{d}(m)$ for $m = n^{0.40}, \ldots, n^{0.85}$.
  3. Compute the Qu statistic with $m = n^{0.7}$ and $\varepsilon = 0.02$.
  4. Interpretation: is the memory of FTSE 100 volatility genuine, and is it stationary?
- **Report**: a table of nine estimates with SEs, one chart, one statistic, one sentence
- Notebook: section B1

B1: solution (1/2) [Solved]



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B1: solution (2/2) [Solved]

m	GPH (SE)	LW	ELW	SE (LW, ELW)
74	0.604 (0.075)	0.601	0.639	0.058
177	0.611 (0.048)	0.578	0.603	0.038
419	0.620 (0.031)	0.606	0.618	0.024

- Qu: $W = 0.33$ against the 10% value 1.09: no evidence of level shifts
- Interpretation: $\hat{d} \approx 0.6$ at every bandwidth with a flat $\hat{d}(m)$: genuine long memory, at or just beyond the stationarity border; report ELW

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B2: BET and EUR/RON with the Qu test [Proposed]

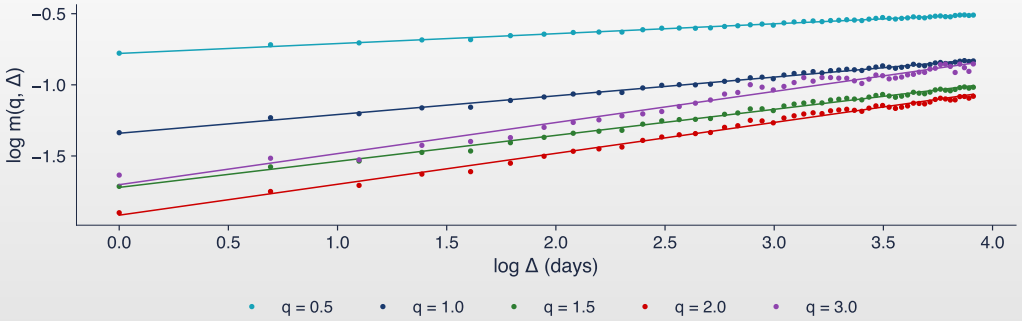
- **Question**: is the memory of $|r_t|$ on the Bucharest market and for the leu genuine, or does it come from regime changes?
- **Inputs**: BET 2000–2026 and from 2010; EUR/RON (BNR) 2005–2026 and from 2014; model: B1
- **Tasks**
  1. Estimate d of $|r_t|$ by local Whittle with $m = n^{\mathbf{0.65}}$ on the full sample and after the cut date.
  2. Compute the Qu statistic ($m = n^{\mathbf{0.7}}$, $\varepsilon = 0.02$) on the same samples.
  3. Interpretation: which of the two markets has genuine long memory in volatility?
- **Report**: four estimates with SEs, four statistics, one sentence
- Notebook: section B2

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B3: roughness of the Euro Stoxx 50 [Solved]

- **Question**: does the Gatheral–Jaisson–Rosenbaum scaling hold for the Euro Stoxx 50, and how rough is its volatility?
- **Inputs**: Oxford-Man 5-minute RV, $n = 5\,635$ days; $\sigma_t = \sqrt{RV_t}$; $q = 0.5, 1, 1.5, 2, 3$; $\Delta = 1, \ldots, 50$
- **Tasks**
  1. Compute $m(q, \Delta)$ and the slopes ζ_q; estimate H by regressing ζ_q on q through the origin.
  2. Estimate H with the measurement-error intercept.
  3. Repeat the first estimate on the two halves of the sample.
  4. Interpretation: is ζ_q linear in q, and what does a departure mean?
- **Report**: five slopes, three values of H, ν, one sentence
- Notebook: section B3

B3: solution [Solved]



- ζ_q = 0.069, 0.132, 0.183, 0.217, 0.218; $\hat{H} = 0.093$, $\hat{\nu} = 0.38$; with the error intercept 0.109; halves 0.091 and 0.095
- Interpretation: linear up to $q = 2$, flat between $q = 2$ and 3: large moments are driven by a few extreme days (heavy tails of the increments), so H from $q \leq 2$ is the safer reading; roughness is stable over time

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B4: Bitcoin roughness by sampling interval [Proposed]

- **Question**: does the estimate of H for Bitcoin depend on the sampling interval of the realised variance?
- **Inputs**: Binance RV of Bitcoin from 1-, 5-, 15-, 30- and 60-minute returns, 2018–2026; model: B3
- **Tasks**
  1. Estimate H for each RV by the moment regression and with the error intercept.
  2. Estimate d of $\log RV$ by local Whittle with $m = n^{0.65}$.
  3. Interpretation: which direction does a coarser grid push H, and why?
- **Report**: ten values of H, five values of d, one sentence
- Notebook: section B4

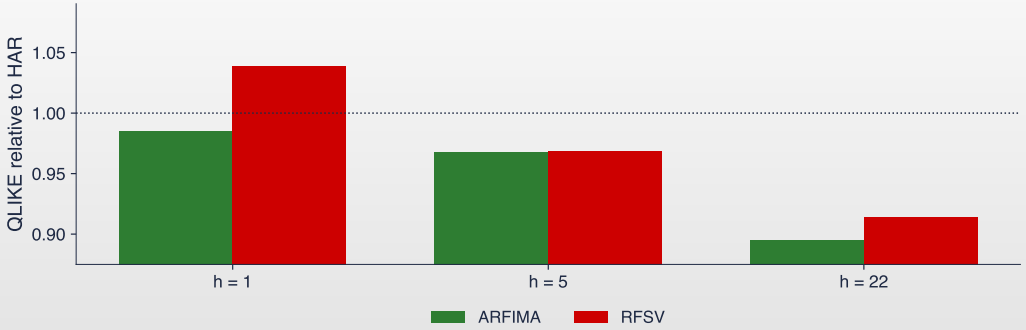
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B5: forecasting FTSE 100 realised variance [Solved]

- **Question**: does a rough or a long-memory model forecast FTSE 100 realised variance better than HAR?
- **Inputs**: rolling window of 1000 days, re-estimation every 20 days; out of sample 30 December 2003 – 26 January 2022 ($T = 4\,565$); horizons 1, 5, 22 days
- **Tasks**
  1. Forecast RV_{t+h} with HAR, ARFIMA$(0, d, 0)$ and RFSV, with the lognormal corrections.
  2. Compute the average QLIKE of each model and the ratio to HAR.
  3. Compute the DM statistics against HAR.
  4. Interpretation: which model would you use at each horizon?
- **Report**: nine losses, six ratios, six DM statistics, one sentence
- Notebook: section B5

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B5: solution (1/2) [Solved]



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B5: solution (2/2) [Solved]

h	QLIKE HAR	ARFIMA / HAR	DM	RFSV / HAR	DM
1	0.230	0.985	−0.98	1.038	1.49
5	0.324	0.968	−1.94	0.968	−1.33
22	0.561	0.895	−2.12	0.914	−1.93

- Interpretation: HAR at one day (RFSV is worse); at 22 days both ARFIMA ($p = 0.034$) and RFSV ($p = 0.053$) beat HAR by about a tenth; median rolling $\hat{H} = 0.11$

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B6: forecasting CAC 40 realised variance [Proposed]

- **Question**: does the FTSE 100 ranking carry over to the CAC 40?
- **Inputs**: the same design, out of sample 22 December 2003 – 26 January 2022 ($T = 4\,631$); model: B5
- **Tasks**
  1. Repeat the three forecasts and the QLIKE ratios.
  2. Compute the DM statistics against HAR.
  3. Interpretation: is there a single best model across the two indices?
- **Report**: six ratios, six DM statistics, one sentence
- Notebook: section B6

C1: co-memory of DAX and CAC 40 volatility [Proposed]

- **Question**: do the volatilities of two neighbouring markets share one fractional trend?
- **Inputs**: log 5-minute RV of the DAX and the CAC 40 (Oxford-Man), $n = 5\,600$ common days; FCVAR with $k = 0$; model: B1
- **Tasks**
  1. Write a five-line pre-registration: hypothesis, model, grid of (d, b), tests, what would change your mind.
  2. Estimate d of each series and of the spread $\beta' X_t$ by local Whittle, then the FCVAR by profile likelihood with trace tests.
  3. Interpretation: what do the data support, and what would you add to the model?
- **Report**: a five-line plan, five estimates, two tests, a project plan
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise long memory and rough volatility. The answer:
- (a) An ARFIMA with d = 0.3 has autocorrelations that decay exponentially, only more slowly than an AR(1).
- (b) The standard error of local Whittle is pi/sqrt(24m).
- (c) A larger bandwidth m is always better, because the variance falls.
- (d) If the Qu test rejects, the series has structural breaks.
- (e) H = 0.1 means that volatility has short memory, because H < 0.5 is antipersistence.
- (f) FIGARCH with 0 < d < 1 is covariance stationary with a finite unconditional variance.
- (g) Exact local Whittle remains consistent for d = 1.2, whereas local Whittle does not.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- Long memory makes i.i.d. and HAC standard errors far too small
- Local Whittle is the default; exact local Whittle when d may exceed 1/2; always vary m
- Test against level shifts before claiming long memory
- Log volatility is rough ($H \approx 0.1$) and persistent; the proxy affects H
- Forecast gains over HAR are small and horizon-dependent

After the seminar

- Lecture 10 develops each topic of today: spectral theory, asymptotics of the estimators, the Qu test, FCVAR, FIGARCH, rough volatility and its forecasts
- C1 can grow into a team project: co-memory of Central and Eastern European volatilities
- Reading: Robinson (1995b); Shimotsu and Phillips (2005); Qu (2011); Gatheral, Jaisson and Rosenbaum (2018)
- **The seminar is for practice and is not graded**

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